

Verifying the Kernel Result on the Report-13 Zoom Points

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Binary: `binary-v5-affinity (fc33e29)`. Tool: Nsight Compute 2026.2. Machine: i7-9750H + GTX 1660 Ti Max-Q (Turing TU116), AC power. Data + capture script: `experiments/17-ncu-zoom-points/`. The four targets are byte-identical to `13-zoom-points.tex`; only the frame count is reduced ($100 \rightarrow 24$).

Resumen

Report 16 measured, on synthetic specs, that the coherent regions GPU affinity routes to the device run divergence-free and FP64-bound. This report verifies that on the four `13-zoom-points.tex` geometries (outside, inside, Misiurewicz, seahorse) — the real workload shapes — and adds Misiurewicz, which report 16 never covered. Same `ncu` methodology: GPU-only capture so the device sees every region type; the same four report-13 targets, with frame count reduced $100 \rightarrow 24$ (warp and FP64 metrics are rates, independent of frame count). The result reproduces exactly: leaves of 480×270 and larger run at 99.7–100 % warp execution efficiency in every regime (0 % below 70 %), while divergence (down to 38 %) is entirely confined to the 240×135 pixel-floor leaves (mean $\sim 89\%$, 8–14 % below 70 %). The inside point is FP64-bound at 85.6 % of peak (~ 110 of 129 GFLOPS at 1.34 GHz), matching report 16 to the decimal. Even Misiurewicz’s thin filaments obey the rule: the filament-crossing floor leaves diverge, the big regions do not. Divergence is confined to the floor tier by the subdivision rule itself — independent of geometry.

1. Setup

What is being verified.

`16-ncu-divergence.tex` established, from Nsight Compute counters, that (i) the coherent interior/exterior regions run at exactly 100 % warp execution efficiency, (ii) the kernel is FP64-bound at $\sim 85\%$ of the 1:32 double-precision peak, and (iii) divergence is real but confined to the pixel-floor boundary leaves. Those measurements used synthetic specs. The open question this report closes: does the result hold on the actual four-regime geometries of `13-zoom-points.tex`, and on Misiurewicz, which report 16 did not include?

The four points.

Identical centres and deep rects to report 13: outside $c=(1,1)$ (escapes at iter 2); inside $c=(-0.5,0)$ (period-1 interior, every deep pixel reaches `maxIter`); Misiurewicz $c=(-0.77568377,0.13646737)$ (self-similar spirals in fast-escape exterior); seahorse $c=(-0.745428,0.113009)$ (filaments + minibrots, the interior-outlier regime).

Frame reduction.

The capture renders 24 frames per spec instead of report 13’s 100. Warp execution efficiency, branch efficiency, and FP64 pipe utilisation are per-kernel rates; they do not depend on how many frames are rendered. The reduction only makes the `ncu` kernel-replay tractable (the heaviest regime, inside, drops from 103 s to 24 s GPU-only). First and last frames, deep-zoom rects, and `maxIter=10000` are unchanged.

Method.

GPU-only (`numThreads=1`) so the single GPU thread processes every region type — in production the affinity queue routes the divergent boundary regions to the CPU pool. Each regime captured twice: `-set full` (roofline / occupancy / stalls / FP64, with warp efficiency) on the 40 largest kernels, and a two-metric warp+branch pass on 300 kernels for the size-resolved distribution. Counters are admin-gated; capture ran as root via `capture.sh`, parsing is unprivileged. GPC clock 1.34 GHz under load.

2. Results

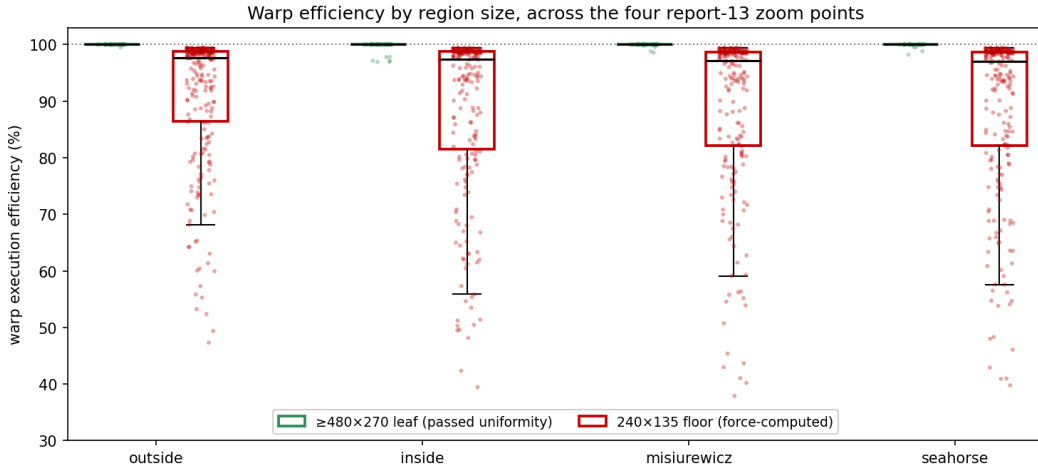


Figure 1: Warp execution efficiency (100 % = all 32 lanes active = zero divergence) by region size, for each of the four report-13 zoom points. In every regime the leaves of 480×270 and larger (green) are pinned at 100 %; only the 240×135 pixel-floor leaves (red) diverge, with a tail to ~ 38 %. The pattern is the same across four geometrically very different targets — divergence tracks region size (floor-clamped or not), not content.

Tabla 1: Per-regime kernel internals (`-set full`, 40 largest kernels). `warpEff` here is over the big kernels; the FP64 pipe and compute Speed-of-Light confirm the kernel is FP64-bound in every regime. Outside is the one exception in cSOL (escapes at iteration 2, so little FP64 work per pixel) — yet still FP64-saturated while it runs.

regime	warpEff %	branch %	FP64 %	cSOL %
outside	100.0	100.0	84.9	81.9
inside	99.7	100.0	85.5	84.1
misiurewicz	99.7	100.0	85.5	84.3
seahorse	99.6	100.0	85.5	84.4

GPC clock 1.33–1.34 GHz; inside FP64 $\approx$ 110 of 129 GFLOPS; L2 hit 92–100 %, memory throughput negligible. Domina

Tabla 2: Warp efficiency split by region size (300-kernel warp pass). The $\geq 480 \times 270$ leaves — the ones that became leaves by passing the uniformity test — never drop below 70 % in any regime; the 240×135 floor leaves carry all the divergence, near-identically across the four geometries.

regime	$\geq 480 \times 270$ leaf		240×135 floor		
	mean %	<70 %	mean %	min %	<70 %
outside	100.0	0 %	91.0	47.3	8 %
inside	99.7	0 %	88.9	39.4	14 %
misiurewicz	99.9	0 %	89.0	37.9	11 %
seahorse	99.9	0 %	88.7	39.8	14 %

3. Analysis

3.1. Coherent regions are divergence-free on every report-13 point

Table 1, Figure 1: across all four geometries the big leaves run at 99.6–100.0 % warp execution efficiency and 100.0 % branch efficiency. Table 2 sharpens it: the $\geq 480 \times 270$ leaves never fall below 70 % in any regime (0 % of them), with a minimum of 96.9 %.

The mechanism is the one report 16 identified, now confirmed on the real geometries: a region only becomes a large leaf if its nine samples passed the uniformity test, and passing means the escape iterations are tight — which is exactly the condition for low intra-warp spread. The uniformity test doubles as a coherence filter, so the leaves that actually run as large kernels are coherent by construction, regardless of whether the target is exterior, interior, a spiral, or a filament field. The verification holds: report 16’s 100 % result was not an artifact of synthetic specs.

3.2. Divergence is confined to the floor tier — the same way in four different geometries

The most striking result is the uniformity of Table 2 and Figure 1. The 240×135 floor leaves diverge in every regime with a mean of 88.7–91.0 %, a minimum of 37.9–47.3 %, and

and 8–14 % of them below 70 %. These numbers are nearly identical across four targets that look nothing alike — a fast-escape exterior, a solid interior, a Misiurewicz spiral, and a seahorse filament field.

The reason is that divergence is a property of subdivision, not of content. A region keeps splitting while its sample spread is high; the only way a high-spread region becomes a leaf is by hitting `pixelSizeThresh` — the floor. So the floor-clamped 240×135 leaves are precisely the regions the uniformity test refused to compute at full size, in every regime. Their internal escape-time spread is what the warp counters see as wasted lanes. The big leaves, having passed the test, are coherent everywhere. The implication is that report 16’s “divergence lives only in the floor leaves” is not regime-specific: it is a structural consequence of the split rule, and it generalises.

Misiurewicz behaves like the rest.

The one regime report 16 never measured is the cleanest test, because a Misiurewicz spiral is mostly fast-escape exterior threaded with thin high-iteration filaments — the classic divergence trap. Yet its big leaves are 99.9 % efficient and only its 240×135 floor leaves diverge (mean 89.0 %, min 37.9 %), exactly like the others. The filaments do cause divergence, but only where they are caught in a floor-clamped leaf; everywhere else the spiral’s coherent exterior dominates a large coherent kernel.

3.3. The FP64 ceiling reproduces to the decimal

Table 1: the FP64 pipe runs at 84.9–85.5 % of peak in every regime, and the inside point — the same geometry as report 13’s heaviest run — is FP64-bound at 85.6 % of peak, ~ 110 of 129 GFLOPS at the measured 1.34 GHz. This is identical to report 16’s interior measurement. Occupancy is 91–95 %, memory throughput is negligible, and the dominant issue stall is `long_scoreboard` at 94–95 % in every regime — the latency of the serial z^2+c recurrence, uniform across content. The FP64-bound, recurrence-latency-limited characterisation of the kernel is reproduced independently on four geometries.

3.4. Why the aggregate looks calmer than report 16

A reconciliation, so the two reports are not misread as conflicting. Report 16’s boundary regime reported 47 % of all kernels below 70 % efficiency; here the worst regime has only 11–14 %. The floor leaves themselves diverge identically (mean ~ 89 %, min ~ 38 % in both reports) — what differs is their fraction of the kernel population. Report 16’s boundary spec was a shallow zoom engineered to maximise boundary content, so floor leaves dominated. The report-13 specs are full zooms whose kernels are mostly coherent (the big leaves plus the 480×270 tier, all 100 %), so the floor leaves are a minority tail. Both agree on the physics of a floor leaf; they differ only in how boundary-heavy the workload is. The production-relevant number is the one here: on real geometries, the GPU’s kernels are coherent at the median (98.5 %) and the divergent floor leaves are both rare and cheap.

4. Implications

1. The report-16 conclusion generalises — divergence is a non-lever. On four real geometries plus Misiurewicz, the GPU’s coherent kernels are 100 % efficient and

the divergence is quarantined to cheap floor leaves that affinity routes to the CPU. There is no warp-divergence optimisation to pursue for the GPU path on any of these regimes.

2. The FP64 pipe is the geometry-independent GPU floor. 85 % of a 1:32 ceiling in every regime, recurrence-latency-limited. The only lever is fewer iterations (work-reduction); `float` is precluded by the deep-zoom precision (report 11). Periodicity checking should still land on the CPU branch first — it would convert these 100 %-efficient kernels into divergent ones (report 16).
3. Floor-leaf divergence is structural, not content-specific. Since it depends only on `pixelSizeThresh` clamping, a smaller `pixelSizeThresh` would shrink the divergent leaves further, but those leaves are already a cheap minority on the CPU — not worth a sweep.

5. Conclusion

Re-running report 16’s Nsight Compute measurement on the exact four `13-zoom-points.tex` geometries reproduces every finding. The coherent leaves GPU affinity routes to the device run at 99.6–100.0 % warp execution efficiency on all four points; the kernel is FP64-bound at 85.6 % of the 1:32 peak (~ 110 of 129 GFLOPS), recurrence-latency-limited; and divergence (to 38 %) is entirely confined to the 240×135 pixel-floor leaves, near-identically across four geometries that look nothing alike — including Misiurewicz, whose filaments diverge only inside floor-clamped leaves. Divergence is a property of the subdivision rule, not of the zoom target; it is a non-lever, and the GPU floor is the FP64 iteration count. The result is verified and generalised.